

A full group of standard-cell descriptions commonly go by the name of ‘technology library.’

Technology libraries are used by commercially available Electronic Design Automation(EDA) tools to automate multiple processes. These include the synthesis, placement and routing of a digital ASIC. It’s the foundry operator that develops and distribute the technology library. The library forms the basis to exchange design information between the various phases of the SPR process.

The phases in the SPR process

Synthesis

The Logic Synthesis tool uses the technology library’s cell logic view to mathematically transform the register-transfer level(RTL) description of the ASIC into a technology-dependent netlist. This is so much similar to how a software compiler would transform a high-level C-program listing into a processor-dependent assembly-language listing.



It even looks like a brick and mortar library's blueprint!

The netlist forms the ASIC design’s standard-cell representation at the level of the logical view. Aside from instances of the standard-cell library gates, it also contains port connectivity between gates. Use of the right synthesis techniques would ensure that the mathematical equivalency exists between the synthesized netlist as well as original RTL description. No unmapped RTL statements and declarations could be found in the netlist.